Min Woo Park, Ph.D.

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SHORT BIO. My research focuses on causal decision making (NeurIPS-25, Preprint, ICLR-26) and causal effect identification (NeurIPS-24). I investigate how to formally model diverse real-world problems within a causal framework and derive optimal decisions from such structured representations. In particular, I develop methods for robust decision-making in settings under environmental uncertainty (NeurIPS-25), domain shifts between training and deployment environments (Preprint), and even scenarios involving counterfactual actions (ICLR-26).

EDUCATION

Graduate School of Data Science, Seoul National University — Seoul, Republic of Korea

Integrated M.S. and Ph.D. in Data Science — Sep. 2022 – Feb. 2027

Advisor: Prof. Sanghack Lee.

Thesis: "Generalizing Causal Decision-Making: Markov Equivalence, Transportability and Counterfactuals".

Overall GPA: 4.02 /4.3; Ph.D. Qualification Exam partially waived due to outstanding academic performance.

Department of Mathematics and Statistics, Sejong University — Seoul, Republic of Korea

B.S. in Applied statistics and Mathematics (Double major), *Cum Laude* — Mar. 2016 – Aug. 2022

Advisors: Prof. Ji Eun Lee and Jin Hee Yoon. — (2.5 years military service)

Overall GPA: 4.04/4.5; Major GPA: 4.46 (Applied Statistics), 4.38 (Mathematics).

PUBLICATIONS

* for joint first authorship and † for corresponding author.

- Min Woo Park and Sanghack Lee†. Counterfactual Structural Causal Bandits. *International Conference on Learning Representations* (ICLR), 2026.
- Min Woo Park, Andy Arditi, Elias Bareinboim†, Sanghack Lee†. Structural Causal Bandits under Markov Equivalence. *Neural Information Processing Systems* (NeurIPS), 2025, Columbia CausalAI Laboratory, Technical Report (R-122)
- Min Woo Park and Sanghack Lee†. On Transportability for Structural Causal Bandits. *arXiv preprint*, 2025. under review.
- Yonghan Jung†, Min Woo Park, and Sanghack Lee†. Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference. *Neural Information Processing Systems* (NeurIPS), 2024.
- Min Woo Park*, and Ji Eun Lee*†. Computation of the iterated Aluthge, Duggal, and Mean transforms. Filomat, 2023.
- Min Woo Park*, Taehui Yun*, Youngin Jang, Younsuk Yeom, Jonghwan Kim, LG AI Research, and Sanghack Lee†. Breaking Bad: Component-wise Parent Deletion for Score-Based Causal Discovery. under review.
- Yesong Choe, Yeahoon Kwon*, Min Woo Park*, and Sanghack Lee†. Canonical Domain Reduction for Partial Counterfactual Identification. under review.
- Ji Eun Lee*† and Min Woo Park*. Convergence of the iterated mean transforms of a 2×2 matrix. under review.

AWARDS AND HONORS

Graduate Student Instructor (GSI) scholarship — Spring 2024, Spring 2025, Spring 2026

BK21 scholarship — Fall 2024, Fall 2025

First place merit scholarship | Sejong University — Fall 2020, Fall 2021

Fourth place award | The 9th College of Natural Sciences Academic Conference, Sejong University

▹ Mitigating spurious regression in time-series data via Granger causality. — Nov. 2021

Third place award | The 8th College of Natural Sciences Academic Conference, Sejong University

▹ The significance of the Euler's number and its applications. — Nov. 2020

RESEARCH EXPERIENCE

Furiosa AI | AI Research Engineer Internship — Jun. 2026 – Nov. 2026, scheduled

Causality Lab, Seoul National University | Ph.D. Researcher — Jan. 2023 – Feb. 2027

Functional Analysis Lab, Sejong University | Undergraduate Researcher — Jul. 2021 – Dec. 2022

PROJECTS

Towards More Scalable Score-based Causal Discovery | LG AI Research — Sep. 2025 – Aug. 2026

- ▷ Leading a project on scalable score-based causal discovery algorithms. (First-author submission under review.)
  - Developed a novel operator enabling score-based causal discovery algorithms to escape local optima.
  - Investigating parallelization strategies for score-based causal discovery algorithms.

Hyundai NGV Data Boot-Campus 2025 | Hyundai Motor — Jul. 2025 – Sep. 2025

- ▷ Developed high-voltage battery performance prediction technology.

Hyundai NGV Data Boot-Campus 2024 | Hyundai Motor — Jul. 2024 – Sep. 2024

- ▷ Developed an intelligent system for predicting vulnerable regions of automotive body panel stiffness using differential geometrybased automatic curvature analysis.

Deep Reinforcement Learning for Recommendation Systems | Coursework Project — Mar. 2024 – Jun. 2024

- ▷ Developed a recommendation system based on DDPG, leveraging multi-modal item features (video, audio, and text) to improve recommendation performance.
  - Source code available at `github.com/MinwooPark96/ddpgRLRS`.

Optimal Mixed Policies Completeness | Oxford University — Aug. 2023 – Feb. 2024

- ▷ Studied complete characterization of conditions under which the intervention effect of a variable is positive.
  Acknowledged for helpful comments in Toward a Complete Criterion for Value of Information in Insoluble Decision Problems. *Transactions on Machine Learning Research* (TMLR), 2024.

TEACHING ASSISTANTS

Causal Inference for Data Science — Spring 2024, Fall 2024, Spring 2025, Spring 2026

- ▷ Topics included Bayesian networks and graphical causal inference.

Machine Learning and Deep Learning for Data Science II — Fall 2023

- ▷ Covered kernel methods, Gaussian processes, and related topics in machine learning.

Basic Mathematics for Artificial Intelligence (Undergraduate) — Spring 2022

- ▷ Covered linear algebra, basic statistics, and probability theory.

ACADEMIC SERVICES, PROFESSIONAL TALKS AND POSTERS

Counterfactual Structural Causal Bandits

ICLR 2026 poster session | Rio de Janeiro, Brazil — Apr. 2026

Structural Causal Bandits under Markov Equivalence

ExploreCSR poster session | Seoul, Republic of Korea — Dec. 2025

NeurIPS 2025 poster session | San Diego, USA — Dec. 2025

JKAIA 2025, NeurIPS preview session | Seoul, Republic of Korea — Nov. 2025

SNU GSDS student research seminar | Seoul, Republic of Korea — Apr. 2025

Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference

SNU GSDS workshop poster session | Jeju, Republic of Korea — Jul. 2025

NeurIPS 2024 poster session | Vancouver, Canada — Dec. 2024

SNU GSDS student research seminar | Seoul, Republic of Korea — Dec. 2024

Conference Reviewer: 2026 NeurIPS, UAI, ICLR, ICML, **2025** ICML

REFERENCE

Sanghack Lee | Graduate School of Data Science, Seoul National University sanghack@snu.ac.kr
Associate professor

Ji Eun Lee | Department of Mathematics and Statistics, Sejong University jieunlee7@sejong.ac.kr
Associate professor

Jin Hee Yoon | Dept. of AI and Information Technology, Sejong University jin9135@sejong.ac.kr
Associate professor

MENTORING

Taehui Yun | Graduate School of Data Science, Seoul National University dbs4835@snu.ac.kr
Ph.D. student. Co-advised a project; paper submitted to UAI 2026.

Last updated: Mar. 16, 2026